

SF1811 Optimization

Lecture 2: Basic feasible solutions

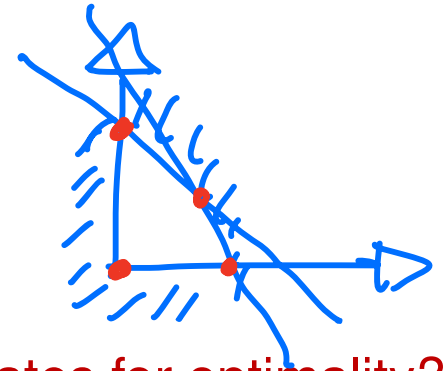
Chapter 4

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Basic feasible solutions

Vertices

Extreme points



Main question:

Why are basic feasible solutions the relevant candidates for optimality?

Today:

- Basic and nonbasic variables.
- Basic solutions.
- Basic feasible solutions.
- Basic feasible solutions and vertices.
- The fundamental theorem of linear programming.

Linear programming in standard form

Consider a linear program in standard form:

$$\begin{aligned} \text{minimize} \quad & c^T x \\ \text{subject to} \quad & Ax = b, \\ & x \geq 0, \end{aligned}$$

$$c^T x = \sum_{j=1}^{n} c_j x_j$$

$$Ax = \sum_{j=1}^{n} a_j x_j$$

a_j: jth column of A

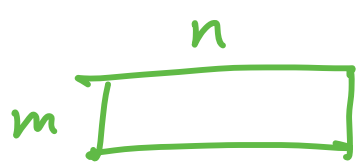
where

$$A \in \mathbb{R}^{m \times n}, \qquad b \in \mathbb{R}^m, \qquad c, x \in \mathbb{R}^n.$$

Assume that

Each row has new information

$$\text{rank}(A) = m, \qquad m < n.$$



The m equations do not determine all n variables uniquely.

Choosing the nonbasic variables



Let $N = (N_1, \ldots, N_{n-m})$ denote a tuple of $n - m$ of indices. The corresponding variables, collected in x_N, are set equal to zero:

$$x_N = 0.$$

Let $B = (B_1, \ldots, B_m)$ denote a tuple of the remaining indices, and let x_B denote the vector of corresponding variable values.

Correspondingly,

$$A = \begin{pmatrix} A_B & A_N \end{pmatrix}, \qquad Ax = A_B x_B + A_N x_N.$$

As $x_N = 0$, the remaining variables x_B must satisfy

$$A_B x_B = b.$$

If A_B is nonsingular, B is called a basis and x_B are the basic variables.

We choose the nonbasic variables. The equations determine the basic variables.

Basis and basic solution

Let B denote a basis and let N denote the tuple of nonbasic indices. The corresponding columns are collected in

$$A_B = \begin{pmatrix} a_{B_1} & \cdots & a_{B_m} \end{pmatrix}, \qquad A_N = \begin{pmatrix} a_{N_1} & \cdots & a_{N_{n-m}} \end{pmatrix}.$$

The solution defined by

$$x_N = 0, \qquad A_B x_B = b,$$

is the basic solution associated with the basis B.

Basic solution and basic feasible solution

A basic solution is defined by

$$x_N = 0, \qquad A_B x_B = b.$$

$x_B = A_B^{-1} b$

$x_N = 0$

$x = \begin{pmatrix} x_B \\ x_N \end{pmatrix} = \begin{pmatrix} A_B^{-1} b \\ 0 \end{pmatrix}$

It is a basic feasible solution if

$$x_B \geq 0.$$

Basic solution

Step	Operation	Question
Choose	Set $x_N = 0$	Does A_B form a basis?
Solve	Solve $A_B x_B = b$	What are the basic-variable values?
Check	Inspect x_B	Is $x_B \geq 0$?

Constructing a basic solution is relatively easy. Whether the resulting solution is feasible is known only after solving $A_B x_B = b$.

The Tables and chairs problem

$Ax = b$

Find d such that

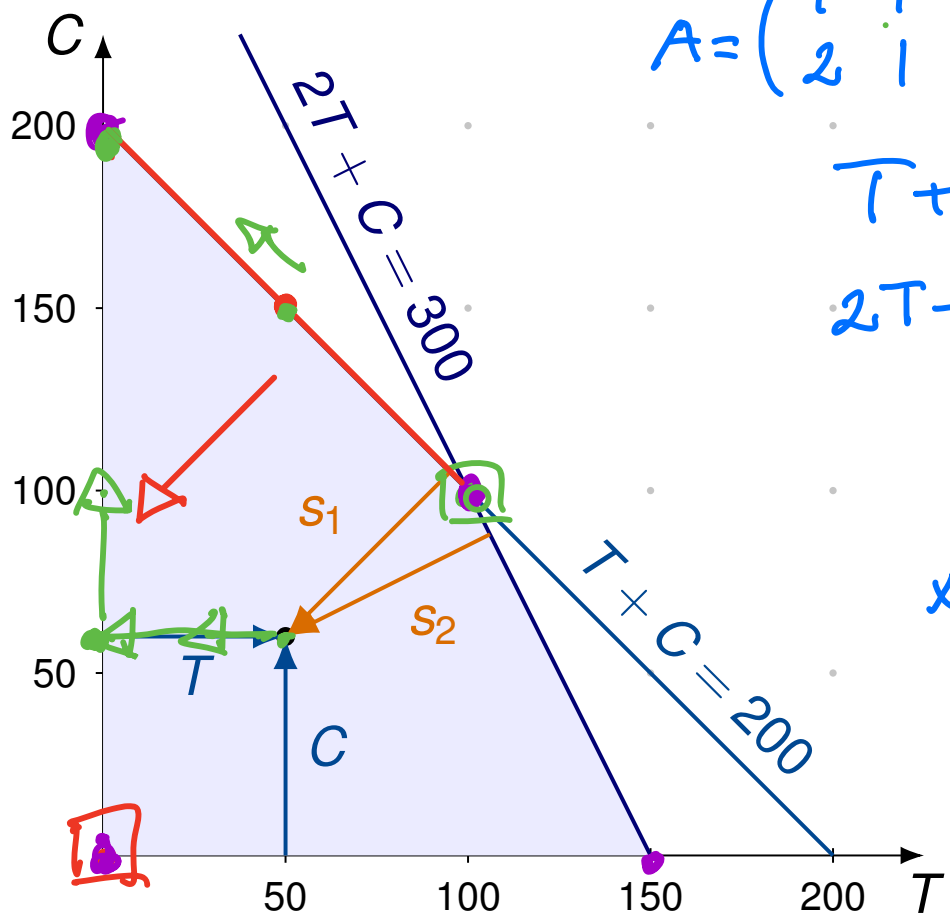
$A(x + td) = b \quad \forall t$

$Ax + tAd = b$

$b \qquad Ad = 0$

$$\begin{pmatrix} T \\ C \\ s_1 \\ s_2 \end{pmatrix} = \begin{pmatrix} 50 \\ 60 \\ 90 \\ 40 \end{pmatrix}$$

(1) $$d = \begin{pmatrix} d_1 \\ d_2 \\ d_3 \\ d_4 \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \\ 1 \\ 2 \end{pmatrix}$$



$$A = \begin{matrix} T & C & s_1 & s_2 \\ \end{matrix} \begin{pmatrix} 1 & 1 & 1 & 0 \\ 2 & 1 & 0 & 1 \end{pmatrix} \qquad b = \begin{pmatrix} 200 \\ 300 \end{pmatrix}$$

$T + C + s_1 = 200$

$2T + C + s_2 = 300$

$$x = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} \qquad x = \begin{pmatrix} T \\ C \\ s_1 \\ s_2 \end{pmatrix}$$

For now

$A(x+td)=b \qquad x+td \geq 0$

$$\begin{pmatrix} T \\ C \\ s_1 \\ s_2 \end{pmatrix} = \begin{pmatrix} 50 \\ 60 \\ 90 \\ 140 \end{pmatrix}$$

$$\begin{pmatrix} 50 \\ 60 \\ 90 \\ 140 \end{pmatrix} + t \begin{pmatrix} -1 \\ 0 \\ 1 \\ 2 \end{pmatrix}$$

$t_{max} = 50$

$$d^{(1)} = \begin{pmatrix} d_1 \\ d_2 \\ d_3 \\ d_4 \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \\ 1 \\ 2 \end{pmatrix}$$

$$\begin{pmatrix} 50 \\ 60 \\ 90 \\ 140 \end{pmatrix} + t_{max} \begin{pmatrix} -1 \\ 0 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 0 \\ 60 \\ 140 \\ 190 \end{pmatrix}$$

$$d^{(2)} = \begin{pmatrix} 0 \\ 1 \\ -1 \\ -1 \end{pmatrix}$$

Tables and chairs: a basic feasible solution

The standard-form constraints are

$$\begin{aligned} T + C + s_1 &= 200, \\ 2T + C + s_2 &= 300, \end{aligned} \qquad T, C, s_1, s_2 \geq 0.$$

Here $m = 2$ and $n = 4$. We therefore choose two nonbasic variables.

Choose

$$T = C = 0.$$

$s_1 = 200 \geq 0$

$s_2 = 300 \geq 0$

Tables and chairs: another basic feasible solution

The standard-form constraints are

$$T + C + s_1 = 200,$$
$$2T + C + s_2 = 300.$$

Choose

$$s_1 = s_2 = 0.$$

$T + C = 200$
$2T + C = 300$

$$\begin{pmatrix} 1 & 1 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} T \\ C \end{pmatrix} = \begin{pmatrix} 200 \\ 300 \end{pmatrix}$$

$$\begin{pmatrix} T \\ C \end{pmatrix} = (-1) \begin{pmatrix} 1 & -1 \\ -2 & 1 \end{pmatrix} \begin{pmatrix} 200 \\ 300 \end{pmatrix} = \begin{pmatrix} 100 \\ 100 \end{pmatrix} \geq \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

Tables and chairs: a basic solution that is infeasible

The standard-form constraints are

$$T + C + s_1 = 200,$$
$$2T + C + s_2 = 300.$$

Choose

$$T = s_2 = 0.$$

$C + s_1 = 200$ $\quad s_1 = -100 \neq 0$

$C = 300$ $\quad C = 300 \geq 0$ Infeasible

Basic solution

Not a basic feasible solution

Basic feasible solutions and vertices

For the Tables and chairs problem:

Basic feasible solution

Nonbasic variables	Basic-variable values	~~Vertex~~
T, C	$s_1 = 200$ $s_2 = 300$	Yes
s_1, s_2	$T = 100$ $C = 100$	Yes
C, s_1	infeasible	–
C, s_2	$T = 150$ $s_1 = 50$	Yes
T, s_1	$C = 200$ $s_2 = 100$	Yes
T, s_2	infeasible	–

The basic feasible solutions obtained algebraically correspond to the vertices found geometrically.

The fundamental theorem of linear programming

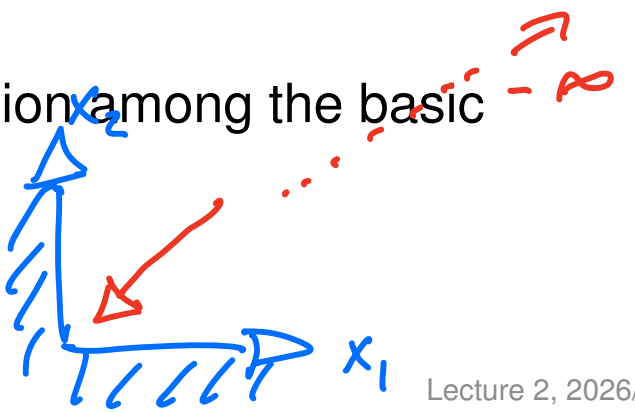


Theorem (The fundamental theorem of linear programming)

Consider a linear program in standard form.

1. *If the linear program has a feasible solution, then it has a basic feasible solution.*
2. *If the linear program has an optimal solution, then it has an optimal basic feasible solution.*

Consequently, it is sufficient to search for an optimal solution among the basic feasible solutions.



Proof idea: feasibility

- Start from a feasible solution x.
- Let $J \coloneqq \{j : x_j > 0\}$.
  If the columns a_j, $j \in J$, are linearly dependent, there is a nonzero vector d such that $Ad = 0$, $d_j = 0$, $j \notin J$.
  After replacing d by $-d$, if necessary, we may assume that d has at least one negative component.
- Hence, $A(x + td) = b$ for every t.
- Choose $t = \min\limits_{j:d_j<0} \dfrac{x_j}{-d_j}$. Then $x + td \geq 0$, and at least one previously positive component becomes zero.
- Repeat until the columns corresponding to the remaining positive components are linearly independent. If necessary, complete these columns to a basis using linearly independent columns corresponding to zero variables.

Proof idea: optimality

x feasible, x optimal

- Now suppose that the starting solution x is optimal.
- Find d as on the previous slide.
- For sufficiently small $t > 0$, both $x + td$ and $x - td$ are feasible.
- Their objective values are

$$c^{\mathsf{T}}(x + td) = c^{\mathsf{T}}x + t\,c^{\mathsf{T}}d, \quad \text{and} \quad c^{\mathsf{T}}(x - td) = c^{\mathsf{T}}x - t\,c^{\mathsf{T}}d.$$

- If $c^{\mathsf{T}}d \neq 0$, one of the two directions improves the objective value. This contradicts optimality. Therefore,

$$c^{\mathsf{T}}d = 0.$$

$c^T(x+td) = c^Tx + t\underbrace{c^Td}_{=0} = c^Tx$

- The reduction can consequently be made without changing the objective value. Repeat as on the previous slide.

Why not enumerate all bases?

n columns
Choose m

There are at most

$$\binom{n}{m} = \frac{n!}{m!(n-m)!}$$

Too big!

ways to choose m columns from an $m \times n$ matrix.

Some choices do not form a basis, and some basic solutions are infeasible.
Nevertheless, the number of possible bases can be very large.

The simplex method searches systematically among basic feasible solutions rather than enumerating all bases.

Main ideas

After this lecture, you should be able to:

- explain the roles of B, N, A_B, x_B, and x_N;
- construct a basic solution by choosing $x_N = 0$;
- determine whether a basic solution is feasible;
- relate basic feasible solutions to vertices;
- explain why an optimal solution can be sought among basic feasible solutions.

Next lecture:

How does the simplex method move from
one basic feasible solution to a better one?